Airway Management

Primary Survey

Airway Compromise Identified

Consider ALS Back-Up

Perform Basic Airway Management
- Clear Airway
- Perform Head-tilt chin lift or Jaw thrust.
- Suction if required

Consider: OPA / NPA — EMR

Support Ventilation
Administer 15L/min Oxygen via NRB mask or BVM as required

Respiratory Monitoring
- SpO2: >94%
- NC-ETCO2: 35-45mmHg

Airway Reassessment
Airway not maintained?
Assess the following:
- Pooling of secretions, blood or vomit in the mouth?
- Unable to swallow?

Impending Airway Failure
- ? FBAO
- Inhalation burns
- Anaphylaxis or Life-threatening Asthma
- Traumatic Brain Injury
- Multiple Traumatic Injuries
- Altered Mental Status unable to maintain airway

Consider CPAP — EMT-A

BVM / Supported Ventilation Rates
- Adult 10-12 b/min
- Child 20-30 b/min
- Infant 20-30 b/min

Continuously monitor ETCO2 for Tidal Volume and Minute Ventilation

Reassess Airway

Consider Advanced Airway Management
Only if patient is in Cardiac Arrest or is deeply unconscious (no gag reflex anticipated)

Consider: Supraglottic Airway (LT) — EMT

Consider: Endotracheal Intubation — P

Advanced Airway Successful?

NO
- Consider Second attempt or/
- Revert to OPA/Basic airway management

YES
Confirm placement:
- ETCO2
- Chest rise/lung sounds
- Sp02

Patient Care Goals

1. Maintain a patent airway.
2. Provide effective oxygenation and adequate ventilation using the most appropriate method to achieve those goals, paired with pulse oximetry and end-tidal capnography (EtCO2) data.
3. Anticipate, recognize, and alleviate respiratory distress.
4. Provide necessary interventions quickly and safely to patients with the need for respiratory support.
5. Anticipate, identify, and plan for a potentially difficult airway.
6. Optimize the patient for any advanced airway attempts, facilitating appropriate initial assessment and management, and linking to specific guidelines as dictated by findings.
7. Achieve a target oxygen saturation of 92-98% to optimize oxygenation and minimize complications associated with hypoventilation.
8. Ensure timely recognition and management of life-threatening conditions (e.g., cardiac arrest, severe trauma) that may require advanced airway techniques.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Airway compromise identified during primary assessment, including:
    - Significantly altered mental status (GCS<8, or unresponsive on AVPU scale)
    - Noisy breathing
    - Stridor
    - Ineffective ventilation
    - Foreign body airway obstruction

Exclusion Criteria

- Patients without identified airway compromise.

Key Patient Assessment Points

1. **Primary Survey**
    - Conduct a primary survey as per Universal Care CPG.
    - Identify airway compromise: Look for signs of obstruction, including noisy breathing, stridor, or ineffective ventilation.
2. **Airway Assessment**
    - Assess the following:
        - Mental status
        - Airway obstruction with foreign body or swelling (e.g., angioedema, posterior pharyngeal and laryngeal infections).
        - Signs of a difficult airway (short jaw or limited jaw thrust or mobility, small thyromental space, upper airway obstruction, large tongue, obesity, large tonsils, large neck, craniofacial abnormalities, excessive facial hair, tracheostomy scar or evidence of other neck/facial surgery, trismus).
3. **Breathing Assessment**
    - Assess the following:
        - Abnormal respiratory pattern, rate and/or effort.
        - Use of accessory muscles.
        - Ability to speak words/sentences.
        - Quality of air exchange, including depth of respiration and equality of breath sounds.
        - Abnormal breath sounds (e.g., wheezing, rhonchi, rales, or stridor).
        - Cough.
        - Skin color (cyanosis or pallor), presence of diaphoresis.
        - Anxiety.
        - Signs of fluid overload (e.g., ascites, peripheral edema).
        - Traumatic injuries impairing upper and lower airway anatomy and physiology:
            - Facial injuries.

- High spine injury (affecting phrenic nerve/intercostals).
- Neck injury (expanding hematoma, tracheal injury).
- Chest wall injury (bruising), including rib and sternal fracture, paradoxical chest motion, subcutaneous air, sucking chest wound.

4. **Airway Reassessment**
   - Airway not maintained? Assess the following:
     - Pooling of secretions, blood, or vomitus in the mouth?
     - Unable to swallow?
   - If unable to maintain airway, consider:
     - Inhalation burns.
     - Anaphylaxis or life-threatening asthma.
     - Traumatic brain injury.
     - Multiple traumatic injuries.
     - Altered mental status unable to maintain airway.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary Survey**
   - Conduct a primary survey as per Universal Care CPG.
2. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
3. **Perform Basic Airway Management**
   - Clear airway: Remove any foreign body, vomitus, or blood using suction or Magill's forceps where indicated – refer to the Foreign Body Airway Obstruction (FBAO) CPG.
   - Attach monitors (SpO2, EtCO2, ECG) and initiate continuous monitoring to assess cardiac rhythm and oxygenation status.
4. **Oxygen Therapy**
   - Administer Oxygen as needed to maintain SpO2 >94%.
5. **Continue Basic Airway Management**
   - Ensure airway patency: Use head-tilt/chin-lift or jaw-thrust maneuvers as appropriate, avoiding neck manipulation if cervical injury is suspected.
   - Clear airway: Remove visible obstructions (e.g., foreign bodies, blood, secretions) using suction or Magill's forceps where indicated – refer to the Foreign Body Airway Obstruction (FBAO) CPG.
   - Maintain airway patency: Use oropharyngeal airway (OPA) or nasopharyngeal airway (NPA) as appropriate, ensuring proper size and avoiding complications (e.g., gagging, bleeding).
   - Provide bag-valve-mask (BVM) ventilation with 100% oxygen if spontaneous breathing is inadequate or ineffective, ensuring a good mask seal and effective ventilation sounds.
6. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - Consider requesting ALS back-up for advanced airway management if:
     - The patient is in cardiac arrest or deeply unconscious (no gag reflex anticipated).
     - Severe respiratory distress or impending respiratory failure.
     - A foreign body is observed in the airway and cannot be removed.
7. **Consider Supraglottic Airway** (LT Airway)
   - Indications:
     - Cardiac Arrest
     - Patient is profoundly unconscious AND basic airway adjuncts (NPA/OPA) fail to maintain airway.
   - Note: EMT's may only insert supraglottic airways (LT Airways) in cardiac arrest.
   - Confirm successful LT airway placement with waveform capnography (EtCO2).
   - Select LT size and LT cuff inflation volume according to colour coded labels on the syringe.

8. **Consider: CPAP / BiPaP**
    - Indications
        - Acute Pulmonary Edema
        - Moderate to severe COPD unresponsive to initial nebulizer treatments.
    - Cautions:
        - Severe pneumonia / ARDS (increase likelihood of pneumothorax)
        - Emphysema (increased pressure may rupture lung bullae)
        - Asthma (may further increase dynamic hyperinflation)
    - Absolute Contraindications
        - Unconscious or altered mental status
        - Apnea
        - Facial Trauma or deformities (unable to gain adequate mask seal)
        - Pneumothorax (untreated)
        - Hypotension
9. **Consider: Endotracheal Intubation**
    - Indications:
        - Cardiac Arrest (see Cardiac Arrest CPG for further guidance on Endotracheal Intubation)
        - Patient is profoundly unconscious AND basic airway adjuncts (NPA/OPA) fail to maintain airway.
    - Ensure equipment and team readiness prior to performing the procedure.
    - If Successful Intubation:
        - Confirm ET tube placement with EtC02 waveform capnography.
        - Connect the bag-valve to the tube and provide ventilations at 1 breath every 6 seconds or as clinically indicated.
        - Connect to a ventilator where available – refer the Invasive Ventilation CPG.
        - Assess airway patency, oxygenation (SpO2), and ventilation (EtCO2) using attached monitors and capnography.
    - If Failed Intubation:
        - Consider a second attempt with ET or supraglottic device, ensuring minimal delay.
            - Ensure optimal airway positioning and vocal cord visualization prior to a second attempt.
    - If the Second Attempt fails:
        - Revert to airway management using supraglottic device (LT airway) or OPA/NPA.
        - Consider consulting Tele-EMS for alternative airway strategies.
10. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
        - **Airway Reassessment**
            - Continuously reassess the patient's airway status, breathing, and response to treatment using attached monitors (SpO2, ECG, EtCO2).
            - Reassess the placement of supraglottic airway (LT) and Endotracheal Tubes after each patient movement. (e.g. from ground to stretcher and from stretcher to hospital bed).
11. **Support Ventilation**
    - Administer 15 L/min oxygen via NRB mask or BVM as required.
        - BVM/Supported Ventilation Rates:
            - Adult: 10-12 b/min.
            - Child: 20-30 b/min.
            - Infant: 20-30 b/min.
            - Continuously monitor EtCO2 for tidal volume and minute ventilation.

Patient Safety Considerations

- Continuously monitor for changes:
    - Use attached monitors (SpO2, EtCO2, ECG) to assess ventilation and oxygenation.

- Detect deterioration using waveform capnography.
- Request ALS back-up for advanced airway management or if unavailable consult Tele-EMS for guidance if basic airway measures fail.

Clinical Insights and Additional Information

- Ineffective ventilation is indicated by the following:

Ineffective Ventilation Signs[3]

Adults	Pediatrics
Inadequate chest rise or minimal movement during breathing attempts.	Inadequate chest rise or minimal movement during breathing attempts, adjusted for smaller size.
Absent or diminished breath sounds on auscultation.	Absent or diminished breath sounds on auscultation, considering smaller airways and higher rates.
Persistent hypoxemia (SpO2 < 92% despite oxygen therapy).	Persistent hypoxemia (SpO2 < 92% despite oxygen therapy, adjusted for paediatric norms).
Increased respiratory effort (e.g., retractions, nasal flaring, use of accessory muscles) without sufficient air movement.	Increased respiratory effort (e.g., retractions, nasal flaring, use of accessory muscles) without sufficient air movement, adjusted for smaller anatomy.
Altered mental status or unresponsiveness due to hypoxia or hypercapnia.	Altered mental status or unresponsiveness due to hypoxia or hypercapnia, considering rapid pediatric changes.
Elevated ETCO2 levels (>43 mmHg) or absence of ETCO2 waveform if capnography is available, indicating poor ventilation.	Elevated ETCO2 levels (>43 mmHg for school-age children, >42 mmHg for infants/toddlers, >40 mmHg for neonates) or absence of ETCO2 waveform if capnography is available, indicating poor ventilation (refer to Special Considerations for pediatric norms).

- Reassess the patient's airway status, breathing, and response to treatment using attached monitors (SpO2, ECG, capnography if available), adjusting for pediatric or adult norms as appropriate.
- Be vigilant for signs of airway obstruction, respiratory distress, or contraindications, especially with airway devices or medications.
- Pediatric considerations: Use age- and size-appropriate equipment, and adjust vital sign norms (e.g., higher heart rate, respiratory rate) to prevent under- or over-treatment. Consult Tele-EMS or call for ALS backup if needed for complex pediatric cases.
- **Pediatric EtCO2 Norms**
  General Range for Healthy Children: In healthy pediatric patients (infants to adolescents), EtCO2 values typically from 30-43 mmHg at sea level, similar to adults but with slight variations based on age, size, and metabolic rate.

Normal Pediatric EtCO2 Ranges

Age Group	EtCO2 Range (mmHg)	Notes
Neonates (0-1 month)	30-40	Values may fluctuate due to higher metabolic rates and smaller tidal volumes.
Infants (1 month-1 year)	32-42	Reflects developing respiratory control and smaller airways.
Toddlers/Preschoolers (1-5 years)	33-43	Stable respiratory patterns mature.
School-Age Children (6-12 years)	35-43	Approaching adult norms as lung capacity and respiratory efficiency increase.

Documentation

- Complete a comprehensive electronic Patient Care Record (ePCR), capturing the following elements to support continuity of care. Including:
    - Identification: Full patient name, age, and gender.
    - Mechanism / Medical Complaint: Presenting symptoms (e.g., noisy breathing, stridor), initial airway status.
    - Injuries / Information: Relevant history (e.g., trauma, burns, allergies).
    - Signs:
        - Full vital signs including pulse, blood pressure, respiratory rate, SpO2, EtCO2.
        - Two sets of vital signs unless critical patient condition precludes.
        - Airway assessment findings (e.g., patency, obstruction).
    - Treatment:
        - Airway management steps and oxygen administration.
        - All interventions: OPA/NPA, BVM, advanced airway attempts, and response.
        - Justification if advanced airway was withheld.
    - Allergies: Documented allergies and drug interactions.
    - Medications: All pre-existing medications and any administered in the field.
    - Background: Relevant medical history, comorbidities, and recent procedures.
    - Other:
        - Communication with Tele-EMS (time and instructions).
        - Response to interventions.
        - Consideration of reversible causes and actions taken.

Key Performance Indicators

1. Document initial airway findings on all patients with airway compromise.
2. Administer oxygen to all patients with SpO_2 <92% (when appropriate).
3. Intubation should be successful on the first intubation attempt.

References

1. Ambulance Tasmania, 2023. *Clinical practice guidelines for ambulance services*. Available at: https://www.ambulance.tas.gov.au (Accessed: 20 June 2025).
2. Ambulance Victoria, 2024. *Clinical practice guidelines*. Available at: https://www.ambulance.vic.gov.au (Accessed: 20 June 2025).
3. American Heart Association, 2020. Guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. Available at: https://www.heart.org (Accessed: 11 July 2025).
4. Benger, J. R., Kirby, K., Black, S., Brett, S. J., Clout, M., Lazaroo, M. J., Nolan, J. P., Reeves, B. C., Robinson, M. and Scott, L. J., 2018. 'Effect of a strategy of a supraglottic airway device vs tracheal intubation during out-of-hospital cardiac arrest on functional outcome: The AIRWAYS-2 randomized clinical trial', *JAMA*, 320(8), pp. 779–791. doi: 10.1001/jama.2018.11597.
5. British Columbia Emergency Health Services, 2024. *BCEHS treatment guidelines*. Available at: https://handbook.bcehs.ca (Accessed: 20 June 2025).
6. Carney, N., Cheney, T., Totten, A. M., Jungbauer, R., Neth, M. R., Weeks, C., Davis-O'Reilly, C., Fu, R., Yu, Y. and Chou, R., 2022. 'Prehospital airway management: A systematic review', *Prehospital Emergency Care*, 26(5), pp. 716–727. doi: 10.1080/10903127.2021.1940400.
7. Deakin, C. D., Clarke, T., Nolan, J., Zideman, D. A., Gwinnutt, C., Moore, F., Ward, M., Keeble, C. and Blancke, W., 2010. 'A critical reassessment of ambulance service airway management in prehospital care', *Emergency Medicine Journal*, 27(3), pp. 226–233. doi: 10.1136/emj.2009.082115.
8. Duckett, J., Fell, P., Han, K., Kimber, C. and Taylor, C., 2014. 'Introduction of the i-gel supraglottic airway device for prehospital airway management in a UK ambulance service', *Emergency Medicine Journal*, 31(6), pp. 505–507. doi: 10.1136/emermed-2012-202126.
9. Frascone, R. J., Russi, C., Lick, C., Conterato, M., Wewerka, S. S., Griffith, K. J., Myers, L., Conners, J. and Salzman, J. G., 2011. 'Comparison of prehospital insertion success rates and time to insertion between standard

endotracheal intubation and a supraglottic airway', *Resuscitation*, 82(12), pp. 1529–1536. doi: 10.1016/j.resuscitation.2011.07.028.

10. Hasegawa, K., Hiraide, A., Chang, Y. and Brown, D. F. M., 2013. 'Association of prehospital advanced airway management with neurologic outcome and survival in patients with out-of-hospital cardiac arrest', *JAMA*, 309(3), pp. 257–266. doi: 10.1001/jama.2012.187612.
11. Hein, C., Owen, H. and Plummer, J., 2008. 'A 12-month audit of laryngeal mask airway (LMA) use in a South Australian ambulance service', *Resuscitation*, 79(2), pp. 219–224. doi: 10.1016/j.resuscitation.2008.06.014.
12. Lockey, D. J., Healey, B., Crewdson, K., Chalk, G., Weaver, A. E. and Davies, G. E., 2015. 'Advanced airway management is necessary in prehospital trauma patients', *British Journal of Anaesthesia*, 114(4), pp. 657–662. doi: 10.1093/bja/aeu412.
13. Middleton, P. M., Simpson, P. M., Thomas, R. E. and Bendall, J. C., 2014. 'Higher insertion success with the i-gel supraglottic airway in out-of-hospital cardiac arrest: A randomised controlled trial', *Resuscitation*, 85(7), pp. 893–897. doi: 10.1016/j.resuscitation.2014.02.021.
14. National Association of EMS Officials (NAEMSO), 2022. *Model EMS clinical guidelines*. Available at: https://naemso.org (Accessed: 20 June 2025).
15. NSW Ambulance, 2023. *Clinical practice guidelines*. Available at: https://www.ambulance.nsw.gov.au (Accessed: 20 June 2025).
16. Queensland Ambulance Service, 2024. *Clinical practice guidelines*. Available at: https://www.ambulance.qld.gov.au (Accessed: 20 June 2025).
17. South Australian Ambulance Service, 2023. *Clinical practice guidelines*. Available at: https://www.saambulance.com.au (Accessed: 20 June 2025).
18. St John Ambulance Western Australia, 2023. *Clinical practice guidelines*. Available at: https://clinical.stjohnwa.com.au (Accessed: 20 June 2025).
19. St John New Zealand, 2024. *Clinical practice guidelines*. Available at: https://www.stjohn.org.nz (Accessed: 20 June 2025).
20. Wang, H. E., Schmicker, R. H., Daya, M. R., Stephens, S. W., Idris, A. H., Carlson, J. N., Colella, M. R., Herren, H., Hansen, M. and Richmond, N. J., 2018. 'Effect of a strategy of initial laryngeal tube insertion vs endotracheal intubation on 72-hour survival in adults with out-of-hospital cardiac arrest: A randomized clinical trial', *JAMA*, 320(8), pp. 769–778. doi: 10.1001/jama.2018.7044.
21. Wiese, C. H., Semmel, T., Muller, J. U., Bahr, J., Ocker, H. and Graf, B. M., 2009. 'The use of the laryngeal tube disposable (LT-D) by paramedics during out-of-hospital resuscitation: An observational study concerning ERC guidelines 2005', *Resuscitation*, 80(2), pp. 194–198. doi: 10.1016/j.resuscitation.2008.10.009.